

concordance

August 27, 2025

1 Concordance and Performance of IVFPQ vs. KNN

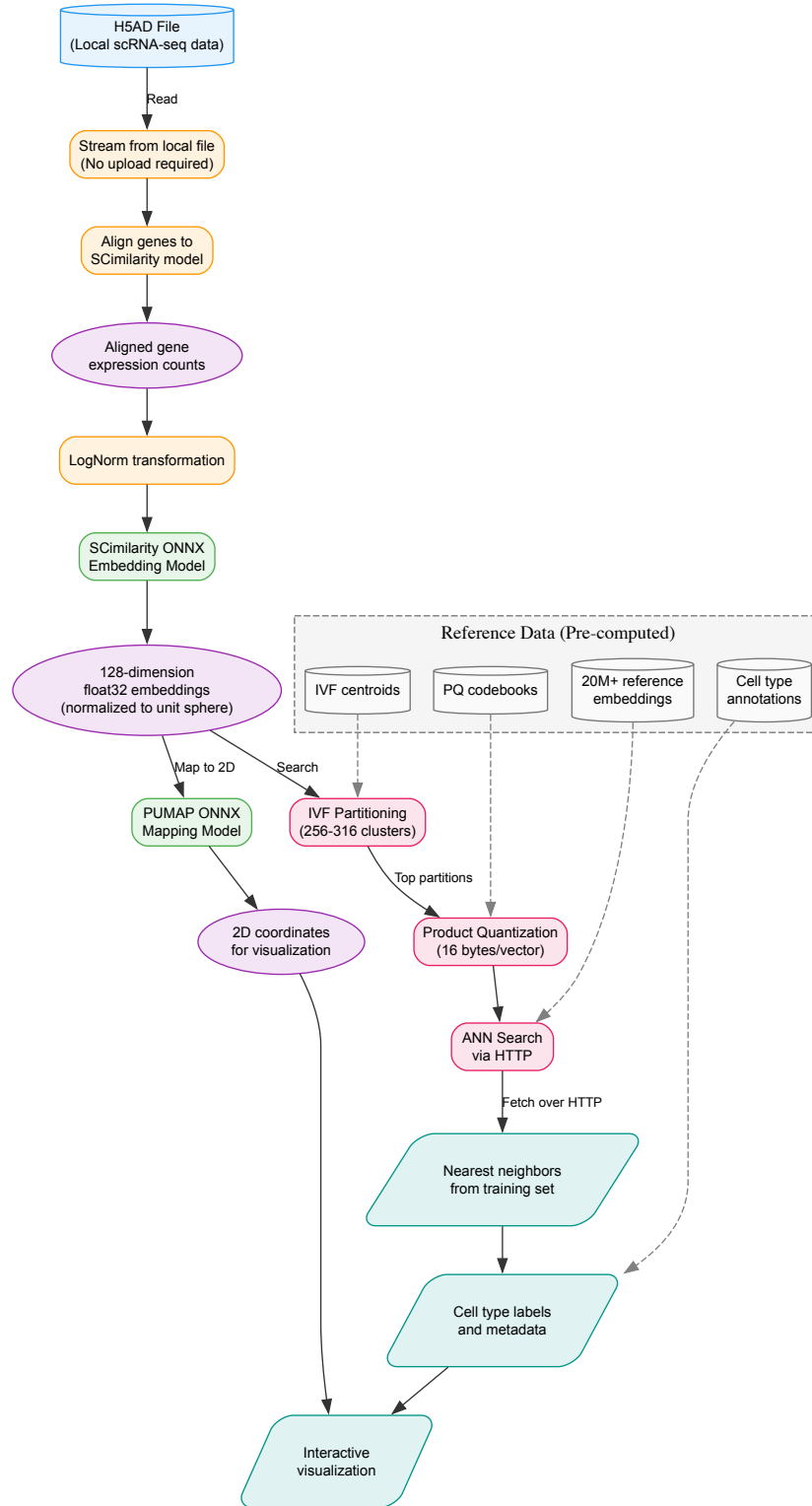
SCimilarity uses TileDB KNN IVF_FLAT index which provides exact vector similarity search.

Cytoverse quantizes the vectors via Product Quantization (PQ) and then performs an approximate lookup based on partitions in an Inverted File Index (IVF) fetched over HTTP.

Note: The memory calculations use `tracemalloc.test_traced_memory()` and are only valid from a fresh run of this notebook on a restarted kernel. We call `gc.collect()` to attempt a clean measurement between each set of parameters

Loaded test data shape: (100, 44942)

1.1 Data Flow



2 Pytorch vs. ONNX Embeddings Concordance

Confirmation that SCimilarity embeddings via PyTorch are identical to those generated via an ONNX exported model.

Max difference between SCimilarity (PyTorch) and ONNX: 3.31e-06

KNN execution time: 35.7604 seconds

KNN memory (~bandwidth) usage: 1056.35 MiB

KNN peak memory usage: 1056.35 MiB

KNN on-disk storage: 9023.19 MiB

IVF partitions: 0% | 0/3 [00:00<?, ?it/s]

Probes (n_sub=8): 0% | 0/8 [00:00<?, ?it/s]

Probes (n_sub=16): 0% | 0/8 [00:00<?, ?it/s]

Probes (n_sub=32): 0% | 0/8 [00:00<?, ?it/s]

2.1 Label Mismatch Analysis

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COMPARATIVE ANALYSIS: Parameter Extremes

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LOWEST PARAMETERS	HIGHEST PARAMETERS
n_sub=8, n_probe=1	n_sub=32, n_probe=32

Mismatch Rate: 25.0%	Mismatch Rate: 2.0%
Total Mismatches: 25	Total Mismatches: 2
Improvement: --	Improvement: 23.0% points

Mismatch Category Distribution

Same Lineage	11 (44.0%)	1 (50.0%)	Δ: +6.0%	
Related	0 (0.0%)	0 (0.0%)	Δ: +0.0%	
Different Immune	13 (52.0%)	1 (50.0%)	Δ: -2.0%	
Cross-Domain	1 (4.0%)	0 (0.0%)	Δ: -4.0%	
Unrelated	0 (0.0%)	0 (0.0%)	Δ: +0.0%	

Query Resolution Analysis

Queries fixed with higher parameters: 24
New mismatches with higher parameters: 1
Persistent mismatches: 1

Persistent Mismatches (still wrong with max parameters)

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Query 67:

TRUE: macrophage
 LOW: alveolar macrophage (confidence: 100%)
 HIGH: alveolar macrophage (confidence: 50%)
 Status: Same wrong prediction

Top 5 Confusions Comparison

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LOWEST PARAMETERS	HIGHEST PARAMETERS

5x alveolar macrop macrophage	1x alveolar macrop macrophage
5x macrophage myeloid cell	1x endothelial cel mesothelial cel
3x alveolar macrop myeloid cell	
2x fibroblast myofibroblast c	
2x alveolar macrop monocyte	

===== DETAILED ANALYSIS FOR EACH CONFIGURATION =====

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 Configuration: n_sub=16, n_probe=4
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Total Mismatches: 8/100 queries (8.0%)

Mismatch Categories:

 Same Lineage : 4 (50.0%)
 Different Immune : 4 (50.0%)

Top 5 Most Frequent Confusions:

 4x: alveolar macrophage macrophage
 [Same Lineage] Conf diff: +0%
 1x: kidney collecting duct pr respiratory epithelial ce
 [Different Immune] Conf diff: +0%
 1x: macrophage myeloid cell
 [Different Immune] Conf diff: +10%
 1x: endothelial cell mesothelial cell
 [Different Immune] Conf diff: +10%
 1x: alveolar macrophage myeloid cell

[Different Immune] Conf diff: -10%

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Configuration: n_sub=16, n_probe=8
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Total Mismatches: 7/100 queries (7.0%)

Mismatch Categories:

Same Lineage : 3 (42.9%)
Different Immune : 4 (57.1%)

Top 5 Most Frequent Confusions:

3x: alveolar macrophage macrophage
[Same Lineage] Conf diff: -3%
1x: kidney collecting duct pr respiratory epithelial ce
[Different Immune] Conf diff: +0%
1x: alveolar macrophage myeloid cell
[Different Immune] Conf diff: +10%
1x: endothelial cell mesothelial cell
[Different Immune] Conf diff: +10%
1x: macrophage myeloid cell
[Different Immune] Conf diff: +20%

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Configuration: n_sub=32, n_probe=4
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Total Mismatches: 8/100 queries (8.0%)

Mismatch Categories:

Same Lineage : 3 (37.5%)
Different Immune : 5 (62.5%)

Top 5 Most Frequent Confusions:

3x: alveolar macrophage macrophage
[Same Lineage] Conf diff: -10%
2x: alveolar macrophage monocyte
[Different Immune] Conf diff: +30%
1x: kidney collecting duct pr respiratory epithelial ce
[Different Immune] Conf diff: +0%
1x: endothelial cell mesothelial cell
[Different Immune] Conf diff: +0%
1x: alveolar macrophage myeloid cell

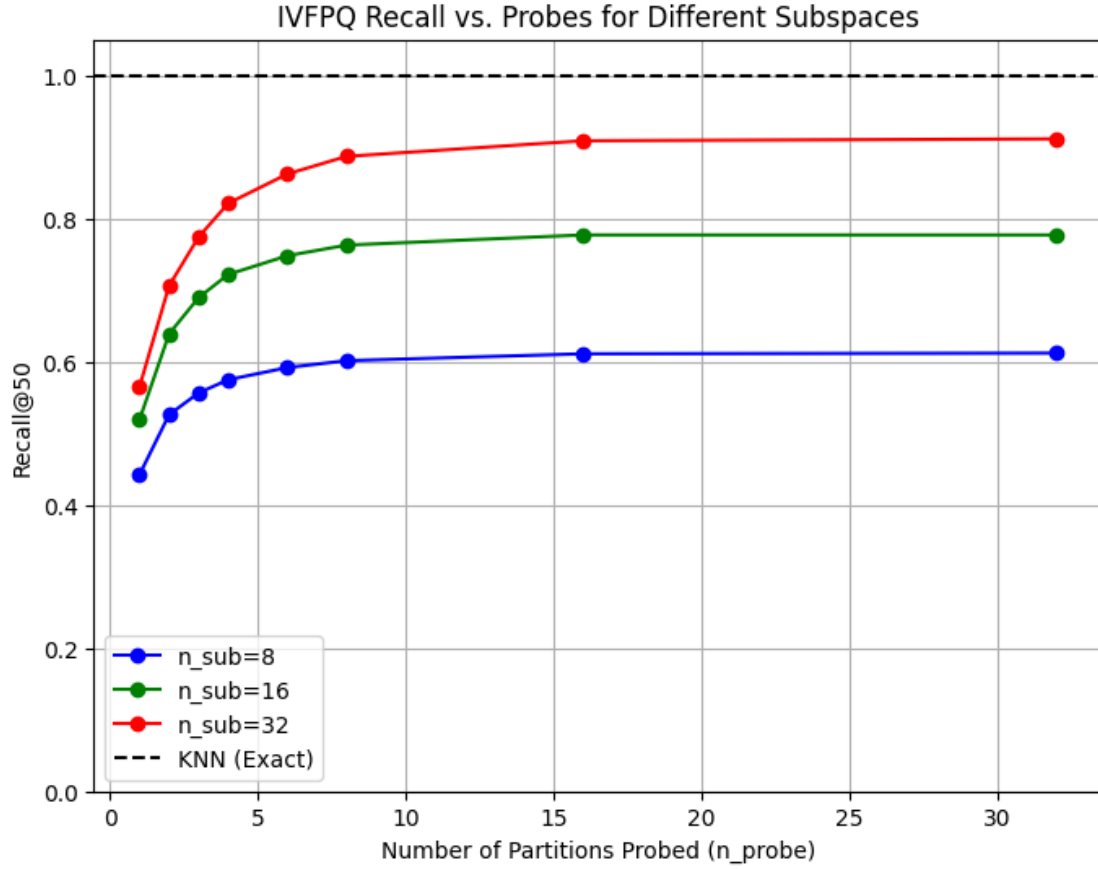
[Different Immune] Conf diff: -10%

2.2 Summary

Method	n_sub	n_probe	Recall	Time (s)	Speedup	Storage	Reduction	BW
IVFPQ	8	1	0.442	1.6	22.8×	92 MB	98×	1.2 MB
IVFPQ	8	2	0.526	3.7	9.7×	92 MB	98×	2.3 MB
IVFPQ	8	3	0.556	5.4	6.6×	92 MB	98×	3.3 MB
IVFPQ	8	4	0.575	7.4	4.8×	92 MB	98×	4.4 MB
IVFPQ	8	6	0.592	11.9	3.0×	92 MB	98×	6.7 MB
IVFPQ	8	8	0.601	16.2	2.2×	92 MB	98×	8.4 MB
IVFPQ	8	16	0.611	33.5	1.1×	92 MB	98×	15.2 MB
IVFPQ	8	32	0.612	72.7	0.5×	92 MB	98×	26.0 MB
IVFPQ	16	1	0.519	1.6	22.2×	153 MB	59×	2.0 MB
IVFPQ	16	2	0.639	3.8	9.4×	153 MB	59×	3.7 MB
IVFPQ	16	3	0.689	5.7	6.3×	153 MB	59×	5.5 MB
IVFPQ	16	4	0.722	7.9	4.5×	153 MB	59×	7.4 MB
IVFPQ	16	6	0.748	12.2	2.9×	153 MB	59×	11.1 MB
IVFPQ	16	8	0.763	16.4	2.2×	153 MB	59×	14.1 MB
IVFPQ	16	16	0.777	33.7	1.1×	153 MB	59×	25.2 MB
IVFPQ	16	32	0.777	70.2	0.5×	153 MB	59×	43.4 MB
IVFPQ	32	1	0.565	2.1	16.6×	274 MB	33×	3.7 MB
IVFPQ	32	2	0.706	4.3	8.3×	274 MB	33×	6.8 MB
IVFPQ	32	3	0.774	6.8	5.2×	274 MB	33×	10.0 MB
IVFPQ	32	4	0.821	8.4	4.3×	274 MB	33×	13.3 MB
IVFPQ	32	6	0.862	12.9	2.8×	274 MB	33×	20.1 MB
IVFPQ	32	8	0.887	18.2	2.0×	274 MB	33×	25.5 MB
IVFPQ	32	16	0.908	37.5	1.0×	274 MB	33×	45.5 MB
IVFPQ	32	32	0.911	74.4	0.5×	274 MB	33×	77.9 MB
KNN	-	-	1.000	35.8	1.0×	9023 MB	1.0×	0.0 MB

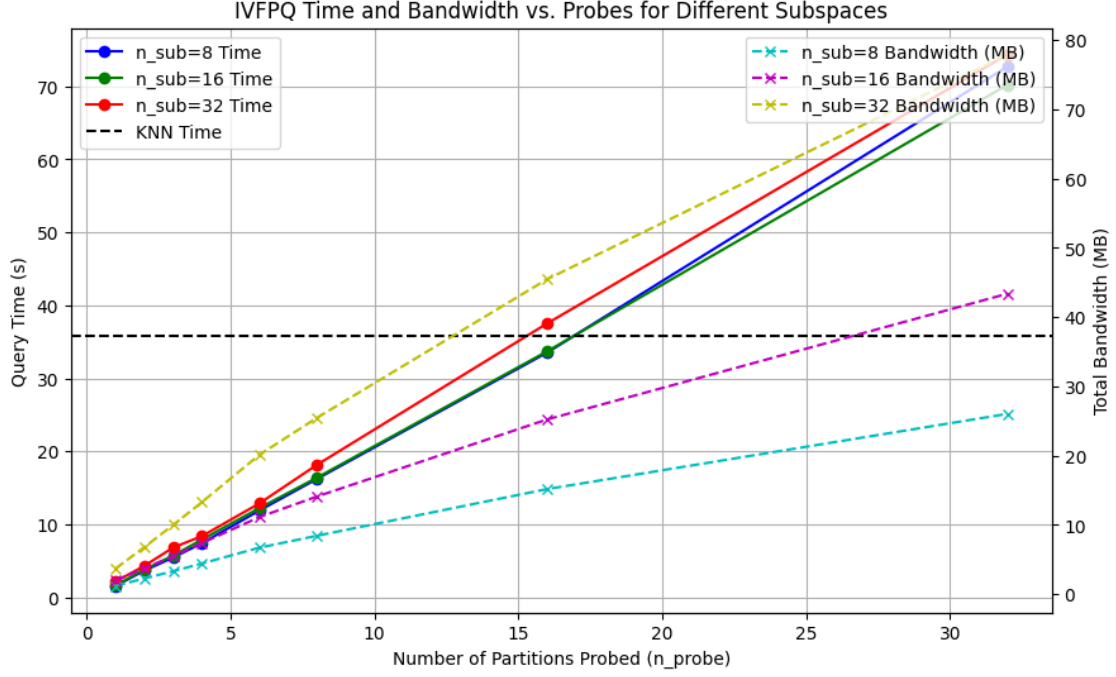
2.3 Recall vs. Probes

IVFPQ recall improves with more probes but plateaus. Fewer subspaces trade accuracy for compression reducing server storage and bandwidth.



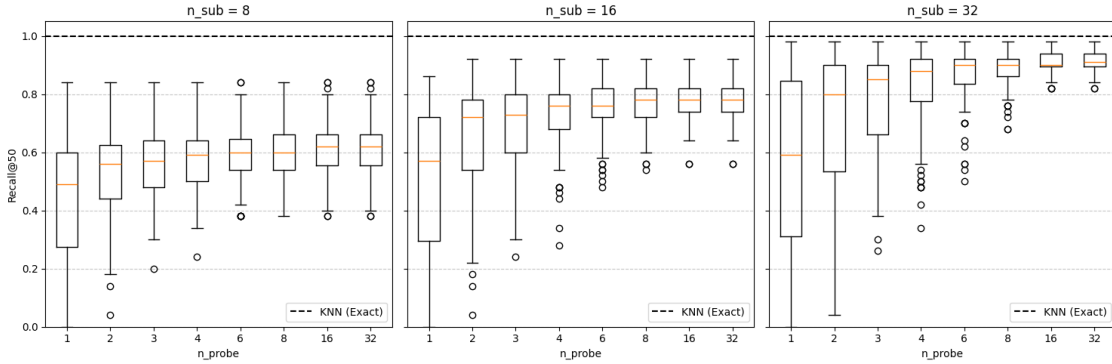
2.4 Time and Bandwidth vs. Probes

Shows tradeoff between PQ quantization and its impact on overall bandwidth and time vs. number of partitions probed. IVFPQ queries are faster than KNN, with bandwidth scaling linearly with probes



2.5 Recall@K by IVGPQ Parameter

Distribution of Recall@K across all queries for each n_probe value grouped by n_sub . Shows variability in recall (not just the mean) across queries, helping researchers assess consistency for single-cell RNA-seq labeling in a web app vs. local KNN.



2.6 Distance Distortion vs. Recall@K

Scatter plots of the relationship between mean distance distortion (for matched neighbors) and Recall@K for each query, with points colored by the average nearest neighbor distance in single-cell RNA-seq data (nn_dist_SC , reflecting query “difficulty”). This helps researchers understand if

queries with farther neighbors (harder cases) exhibit higher distortion or lower recall in IVFPQ compared to KNN, relevant for web app vs. local tool tradeoffs.

Distance Distortion vs. Recall@K (n_probe horizontal, n_sub vertical)

